PROJECT SUMMARY

ReqCheck

Software Requirements Review & Classification

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Project objective

Develop a human-in-the-loop requirements workspace that helps reviewers import specifications, repair extracted statements, explore functional/non-functional classification and quality cues, and record traceable decisions. A separate empirical study evaluates lightweight methods on held-out source projects.

Objectives

- Import TXT, DOCX and searchable PDF files; edit, add, split, merge and remove requirement statements while retaining source evidence.
- Provide functional/non-functional and exploratory NFR subtype suggestions, explainable ambiguity cues and lexical duplicate candidates.
- Keep decisions reviewer-led, preserve immutable analysis inputs and review history, and export evidence-linked charts and reports.
- Evaluate lightweight TF-IDF and surface-feature classifiers using project-grouped validation, with transparent methods and limitations.

Documented contribution

Conservative, workspace-based summary; not an independent authorship attestation.

- Project specification, architecture, acceptance criteria and implementation plan; research data provenance, cleaning, experiment scripts and archived results.
- React + TypeScript frontend and FastAPI application workflows, trusted model inference, MongoDB storage, requirement repair and reviewer decision history.
- Immutable source snapshots, stale-run detection, four charts, CSV/JSON exports, printable HTML reports, regression tests and real-stack verification.

Implementation status: local core application complete and verified.

Real bundled models and MongoDB integration pass; saved data survives an API restart. Independent external validation, annotated threshold evaluation, usability studies and production hardening remain separate gates.

Stack: React + TypeScript, FastAPI, MongoDB, scikit-learn and Matplotlib.

DOCUMENTED OUTCOMES

Results & evidence

Verified local implementation and carefully stated research results.

Application outcome

17 / 17 tests passed

Verified real model inference and MongoDB integration, repeated snapshots, review validation, exports, four charts, reconnection and cascading deletion.

An actual API restart preserved seven active demo statements, three analysis runs, five reviewer events and the latest edited text. Database and model health remained ready.

TypeScript compilation and browser checks passed: analysis, reviewer decisions, history, historical source isolation, keyboard dialogs, reduced motion and 390px/1440px layouts. These are functional checks, not a usability study.

Research outcome

620 retained requirements from 15 source projects; project-grouped nested cross-validation (five outer folds; up to three inner folds). Values are mean outer-fold macro-F1.

Binary method	Macro-F1
Word TF-IDF + linear SVM	0.6144
Word + surface features + SVM	0.7194
Observed mean difference	+0.1050

Exploratory NFR subtypes: highest observed mean macro-F1 was 0.4752 for word + character SVM across 11 NFR labels.

Interpretation & limits

- Exploratory, retrospective results; data cleaning followed an initial pilot. The current data mirror is not independently proven byte-identical to a historical archive.
- Fold variability is descriptive, not a confidence interval. Results do not establish statistical significance or external industrial generalization.
- NFR support is sparse, including one portability example. Duplicate threshold 0.75 is provisional. No independent external test, annotation agreement or end-user usability study is available.

Sources: Research_Paper.md; benchmark_results/binary/summary.csv; benchmark_results/nfr/summary.csv; Verification_Report.md. The following original IEE evaluation page supplies the grade-A evidence.

Courses Presented			Grade	Credits
	:	Software Requirement and Specifications	B-	3.00
	:	Object Oriented Software Engineering	A	3.00
	:	Advance Programming Techniques	A	3.00
	:	Probability and Statistics	B+	3.00
Semester V	:	Business and Technical Writing	A+	3.00
	:	Software Engineering Economics	B+	3.00
	:	Computer Communications and Networks	B+	4.00
	:	Data Warehousing and Data Mining	B+	3.00
	:	Software Design and Architecture	B+	3.00
Semester VI	:	Software Application Development	B-	3.00
	:	Research Methodology	B+	3.00
	:	Software Verifications and Validation	A	3.00
	:	Formal Methods in Software Engineering	B+	3.00
	:	Introduction to Economics	B+	3.00
	:	Professional Practices	B+	3.00
Semester VII	:	Mobile Application Development	B-	3.00
	:	Software Project Management	B	3.00
	:	Artificial Intelligence	B-	3.00
Semester VIII	:	Final Year Project	A	6.00
	:	Agile Software Engineering	A+	3.00
	:	Research Trends in Software Engineering	B	3.00
	:	Total Quality Management	B+	3.00
			Total	131.00

Grade Point Average is 3.40 based on a 4-point scale with A+/A = 4, A- = 3.67, B+ = 3.33, B = 3, B- = 2.67, C+ = 2.33, C = 2, C- = 1.67, D+ = 1.33, D = 1, D- = 0.67, F = 0.

STATEMENT OF EVALUATION

IEE evaluations and assessments are based on the judgment of evaluators experienced in international education, a review of current literature, and documentation provided. We are members of NACES (National Association of Credential Evaluation Services), AACRAO (American Association of Collegiate Registrars and Admissions Officers), TAICEP: The Association for International Credential Evaluation Professionals, and NAFSA: Association of International Educators. The evaluation methodologies used at IEE include both year-counting and benchmarking—a detailed description of both models is available on our website. Implementation of these is dependent upon the report's purpose and the receiving organization. IEE will prioritize years of full-time study as foundational to the equivalency determination for immigration and licensure purposes, as well as for reports created for specific colleges and universities which have opted out of benchmarked equivalencies. For most education and employment reports, however, IEE will prioritize academic and professional access, curriculum rigor, and contact hours as central to the equivalency determination. This evaluation is simply advisory and is in no way binding on any institution, agency, or organization, each of which has the authority to make decisions that it chooses regarding the application of this analysis.

**********This is the final line of this report, nothing follows **********